

Reliable Evaluation of ARIMA-LSTM Hybrids for Remaining Useful Life Prediction: A Multi-Seed, Specification-Sensitive Study on NASA C-MAPSS

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Abstract—A common design choice for remaining useful life (RUL) prediction is to augment an LSTM with a classical trend feature, typically ARIMA, fit per sensor channel. We ask whether this holds up under multi-seed, multi-specification, statistically corrected evaluation on NASA C-MAPSS, rather than proposing it as a new architecture. We compare a per-channel ARIMA-LSTM hybrid against a plain LSTM, Random Forest, and CNN-LSTM on all four C-MAPSS sub-datasets under six seeds, testing three ARIMA specifications: a fixed order, an AIC-selected order, and a fully causal expanding-window construction that never uses future information. Under the causal, deployable specification, the hybrid shows no Holm-Bonferroni-corrected advantage over plain LSTM on any dataset (closest case FD002, adjusted $p = 0.058$). A decisive control replacing ARIMA with a free, fitting-free causal moving average significantly *outperforms* causal ARIMA on FD002 (adjusted $p = 0.023$) at one-quarter the compute cost. Under the specification used in most prior work, a fixed ARIMA order is significantly worse than plain LSTM on FD004 ($p = 0.004$); AIC-based order selection repairs that disadvantage without beating plain LSTM anywhere. No specification tested yields a corrected-significant hybrid-over-plain win. The contribution is methodological: evaluation choices, seeds, specification, and causal validity, determine what a predictive-maintenance study can defensibly claim.

Index Terms—remaining useful life, predictive maintenance, LSTM, time-series forecasting, feature engineering, industrial IoT, NASA C-MAPSS, reproducibility

I. INTRODUCTION

Estimating remaining useful life (RUL) from streaming sensor data is central to prognostics and health management (PHM), letting maintenance be scheduled proactively [1]. NASA’s C-MAPSS turbofan simulation [1] is the field’s standard benchmark: a controlled, multi-regime, multi-fault-mode setting where model-comparison claims can in principle be checked. A common architectural idea is to fit an ARIMA model per sensor channel and concatenate its trend with the raw signal before an LSTM regressor [10], letting the network spend capacity on residual nonlinear dynamics. But deep models are trained stochastically, and an ARIMA feature is itself a specification choice (order, fitting window, causality), so a single-seed, single-specification claim that this helps cannot be trusted at face value; prior turbofan RUL papers, including

hybrid-forecasting work we cite favorably [6], typically report neither seed count nor causal validity.

This paper evaluates, rather than proposes, ARIMA trend augmentation. We test the hybrid against plain LSTM, Random Forest, and CNN-LSTM baselines across all four C-MAPSS sub-datasets, six seeds each, under three ARIMA specifications, fixed order, AIC-selected order, and a fully causal expanding-window construction, applying Holm-Bonferroni correction consistently across every four-dataset family.

Research questions. RQ1: does a fully causal ARIMA trend feature consistently improve LSTM RUL prediction? RQ2: how sensitive are plain-vs-hybrid comparisons to stochastic training variation? RQ3: does ARIMA order specification materially change the outcome under the protocol used in most prior work? RQ4/RQ5 (combined below): can a much cheaper, fitting-free causal trend summary reproduce whatever effect ARIMA has, and what survives once every comparison is corrected?

Contributions: (1) a multi-seed, four-dataset evaluation of ARIMA-LSTM RUL prediction against matched baselines; (2) a causal feature-construction analysis removing a train/test asymmetry present in the standard full-series ARIMA fit; (3) a decisive control showing a free causal moving-average feature significantly beats causal ARIMA on the one dataset ARIMA looks most promising on; and (4) consistent Holm-Bonferroni correction across every inferential family, publicly released code and per-seed results (full experimental record, including the order-specification sensitivity analysis, window-length ablation, and gated-fusion check, at <https://github.com/PradeepSomasundaram1512/hybrid-arima-lstm-rul-cmapss>).

II. RELATED WORK

The C-MAPSS benchmark [1] underlies most data-driven RUL work; recent architectures include CNN-LSTM [4] and residual-corrected BiLSTM [5] hybrids. ARIMA-neural hybrids have a long history in general forecasting [10] and have been applied to RUL-adjacent problems [6], typically without reporting seed count or causal validity of the ARIMA construction. Separately, reproducibility work outside this subfield

has shown that single-seed deep-learning results routinely fail to replicate [7] and that exact reproducibility does not imply robustness to small setup changes [8]; Demšar [9] sets out the paired, multiple-comparison-corrected testing methodology we follow. A contemporaneous benchmark on this exact dataset [11] uses a single fixed seed, illustrating the gap this paper addresses. Our approach differs in treating ARIMA specification and causal construction as first-class experimental variables, not implementation detail, evaluated across all four regimes and six seeds each.

III. METHODOLOGY

Data and preprocessing. We use all four C-MAPSS sub-datasets [1]: FD001/FD003 (100 engines, single/dual fault mode, 1 condition), FD002/FD004 (260/249 engines, 1/2 fault modes, 6 conditions). RUL is capped at 125 cycles; low-variance sensors are dropped (15 of 21 retained for FD001/FD003). For FD002/FD004, we K -means cluster the three operating-setting dimensions into 6 regimes (fit on training data only) and z -score normalize per regime [3], confirmed seed-stable up to label permutation.

ARIMA trend feature, three specifications. For each (unit, sensor), we fit ARIMA and use its in-sample fitted trend as an extra input channel, concatenated with raw features before a 2-layer LSTM (hidden 64) [2]. *Fixed order* (main comparison): ARIMA(1, 1, 0), parameters fit once via MLE over each unit’s full available series. *AIC-selected order*: the best of four candidate orders under the identical full-series fit, isolating order choice from causality; we label this explicitly as a specification-sensitivity analysis under the non-causal protocol, not causal evidence. *Causal, expanding-window*: ARIMA(1, 1, 0) refit by full MLE every 20 cycles on an expanding window, with Kalman-filter state extension between refits, so no parameter is ever estimated from data at or after the cycle it predicts; this is our primary, deployable specification. Baselines: a plain LSTM (identical architecture, raw features only), a Random Forest (300 trees), and a CNN-LSTM (2-layer 1D-CNN front end, no ARIMA feature) [4]. All models use Adam (10^{-3}), MSE loss on capped RUL, 15% held-out validation engines for early stopping, window length 30 cycles. We report RMSE, MAE, and the asymmetric PHM score [1] (over-late predictions penalized more than early ones), each test engine scored once from its final window.

IV. EXPERIMENTAL SETUP

All runs use six seeds (42, 123, 2024, 7, 99, 314), identically applied to every specification and baseline; PyTorch/statsmodels/scikit-learn, CPU-only. We started with three seeds, found a significant result, and only treated it as reliable after replicating with three more, which changed the answer (Section V). Holm-Bonferroni correction is applied within every four-dataset inferential family in this paper.

V. RESULTS

A. RQ1 – Causal, Deployable Hybrid vs. Plain LSTM

Table I compares plain LSTM against the causal-ARIMA hybrid. The pattern is inconsistent across regimes: worse than

TABLE I
FIXED-ORDER HYBRID RMSE, CAUSAL VS. NON-CAUSAL ARIMA, MEAN $\pm$ STD OVER 6 SEEDS.

Dataset	Plain	Non-causal hybrid	Causal hybrid
FD001	14.43 $\pm$ 0.64	14.92 $\pm$ 1.65	15.39 $\pm$ 1.11
FD002	14.59 $\pm$ 0.49	14.93 $\pm$ 0.60	14.64 $\pm$ 0.36
FD003	14.72 $\pm$ 2.06	14.87 $\pm$ 0.58	14.22$\pm$0.56
FD004	14.92 $\pm$ 0.38	15.71 $\pm$ 0.54	15.59 $\pm$ 0.44

TABLE II
RMSE ACROSS NASA C-MAPSS, MEAN $\pm$ STD OVER 6 SEEDS, NON-CAUSAL PROTOCOL.

Dataset	Plain LSTM	Hybrid ARIMA-LSTM	CNN-LSTM
FD001	14.43$\pm$0.64	14.92 $\pm$ 1.65	14.47 $\pm$ 0.59
FD002	14.59$\pm$0.49	14.93 $\pm$ 0.60	15.11 $\pm$ 1.05
FD003	14.72 $\pm$ 2.06	14.87 $\pm$ 0.58	13.90$\pm$0.70
FD004	14.92$\pm$0.38	15.71 $\pm$ 0.54	14.86 $\pm$ 0.54

plain on FD001 (15.39 vs. 14.43) and FD004 (15.59 vs. 14.92), close on FD002 (14.64 vs. 14.59), better on FD003 (14.22 vs. 14.72). A paired Wilcoxon test on per-engine error, Holm-corrected across the four datasets, finds no significant difference on any dataset (closest: FD002, raw $p = 0.0145$, adjusted $p = 0.058$). **RQ1 answer:** no consistent improvement, and no dataset survives correction.

B. RQ2 – Sensitivity to Stochastic Training Variation

Under the original (non-causal) protocol, Table II shows the plain LSTM’s mean RMSE at or below the hybrid’s on every dataset. This comparison itself moved with seed count: at three seeds, a paired Wilcoxon test found FD001 significant ($p = 0.007$) and nowhere else; adding three more seeds erased FD001’s significance ($p = 0.157$) and made FD004 significant instead ($p = 0.004$, Holm-adjusted, the only survivor across the four-dataset family). Cohen’s d_z and bootstrap 95% CIs confirm only FD004’s interval excludes zero ($d_z = 0.15$, CI [0.11, 1.01]); all effects are small by convention. A CNN-LSTM baseline (no ARIMA feature) shows the same dataset-dependent pattern rather than a uniform result (significantly worse than plain on FD002, $p < 0.001$; non-significantly best on FD003). **RQ2 answer:** which specific dataset a significance claim can defend changed materially between three and six seeds.

C. RQ3 – Order-Specification Sensitivity (Non-Causal Protocol)

Holding the full-series fitting procedure fixed and varying only ARIMA order (four AIC-scored candidates vs. the fixed default) changes the FD004 picture materially: AIC-selected order significantly outperforms fixed order on FD004 (Holm-adjusted $p = 0.0004$), repairing the fixed-order hybrid’s disadvantage from RQ2. Selected order shows its largest numerical gap over plain LSTM on FD002 (13.77 vs. 14.59 RMSE), but this does not survive correction ($p_{\text{adj}} = 0.125$). We label this explicitly as a specification-sensitivity finding

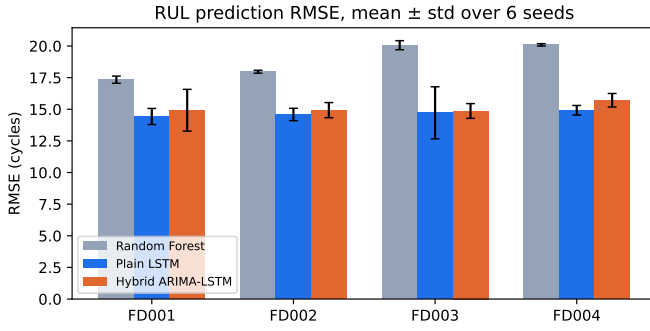


Fig. 1. RMSE across all four sub-datasets, mean $\pm$ 1 std over six seeds, non-causal protocol. Plain LSTM’s mean RMSE is at or below the hybrid’s on every dataset.

TABLE III
CAUSAL TREND-FEATURE CONTROLS, MEAN $\pm$ STD OVER 6 SEEDS.

Dataset	Plain	MA-hybrid	Causal ARIMA
FD001	14.43 $\pm$ 0.64	14.93 $\pm$ 0.86	15.39 $\pm$ 1.11
FD002	14.59 $\pm$ 0.49	14.10 $\pm$ 0.45	14.64 $\pm$ 0.36
FD003	14.72 $\pm$ 2.06	14.87 $\pm$ 0.87	14.22 $\pm$ 0.56
FD004	14.92 $\pm$ 0.38	14.81 $\pm$ 0.45	15.59 $\pm$ 0.44

under the original, non-causal protocol, not evidence that a causally-fit, AIC-selected model would behave the same way (untested). Full order-selection tables, a window-length ablation, a gated-fusion architectural check, and a mechanistic analysis of FD002’s sensor-trend linearity (all further negative or inconclusive results) are reported in full in the repository; none produces a corrected-significant hybrid-over-plain win.

D. RQ4/RQ5 – Decisive Control and Synthesis

If ARIMA’s specific autoregressive structure, not just “some trend feature,” drives its effect, a much cheaper causal alternative should not match it. We replace the causal-ARIMA feature with a trailing 10-cycle moving average (MA) and an exponential moving average (EMA, $\alpha = 0.3$), both fitting-free and trivially causal, under the identical six-seed protocol (Table III). Neither MA nor EMA differs significantly from plain LSTM on any dataset (Holm-adjusted $p \geq 0.91$). Against causal ARIMA, however, MA is significantly *better* on FD002 (raw $p = 0.0058$, Holm-adjusted $p = 0.023$): a mean per-engine error reduction of 0.629 cycles ($d_z = -0.14$, 95% CI $[-1.16, -0.07]$), at one-quarter the wall-clock cost (501s vs. 2103s mean per-seed runtime). This is the paper’s second and final corrected-significant result, and it argues against ARIMA specifically, not trend augmentation broadly.

Across every specification tested and after consistent correction, we find exactly two corrected-significant effects in the paper, and neither is a hybrid-over-plain win: AIC-selected order beats fixed order on FD004 (RQ3), and a free moving average beats causal ARIMA on FD002 (RQ4/RQ5). No comparison of any ARIMA-augmented hybrid against plain LSTM survives correction, under any specification, on any dataset.

VI. DISCUSSION

The paper’s value does not depend on ARIMA winning. A naively fixed order significantly hurts on FD004; correcting the order repairs that specific harm without producing a win; removing the deeper causal asymmetry (RQ1) does not turn any dataset into a corrected-significant win either; and where a trend feature shows its largest promise (FD002), a free rolling mean beats the expensive ARIMA construction outright. Taken together, specification quality and causal validity are separate axes of evaluation, not substitutes for one another, and a single-seed or single-specification claim on this benchmark, or similar small-to-moderate industrial datasets, is not sufficient evidence either way.

VII. LIMITATIONS

Six seeds is a matched, moderate-power design, not exhaustive stochastic validation, though sufficient here to overturn this paper’s own three-seed conclusion twice. All experiments use simulated turbofan data; generalization to real industrial IoT sensor streams is untested. The causal specification costs $3.7\text{--}10.1\times$ the non-causal pipeline’s runtime, a practical deployment consideration we surface but do not optimize. Full limitations, including an unresolved FD003 variance anomaly and an inconclusive mechanistic check of FD002’s sensor-trend linearity, are documented in the repository’s full technical report.

VIII. CONCLUSION

Evaluated across three ARIMA specifications, four C-MAPSS sub-datasets, and six seeds each with consistent Holm-Bonferroni correction, an ARIMA-augmented LSTM shows no corrected-significant advantage over a plain LSTM anywhere, under its causal, deployable construction or otherwise. Order specification and causal validity each materially change individual comparisons without changing this overall conclusion, and a free causal moving-average control significantly outperforms causal ARIMA on the one dataset the hybrid looked most promising on. The methodological point, that evaluation choices determine what a predictive-maintenance study can defensibly claim, is the paper’s contribution, independent of whether this specific hybrid is adopted.

CODE AND DATA AVAILABILITY

Full code, per-seed results, the complete technical report (all research questions, ablations, and negative results reported here in full), and this paper are available at <https://github.com/PradeepSomasundaram1512/hybrid-arima-lstm-rul-cmapss>: 150 pipeline executions across all specifications and baselines, training 360 individual models in total.

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